

Configuration and provenance

Branch: feat/ej230-ssl4

Commits: 0201413, 7b4bc80, ad26d78, bcefa24, bdf955; this deck is generated from those published sources

- ▶ 31 positions $\times$ 2000 events; 16 End + 70 Top channels; jitter fixed to zero.
- ▶ EJ-230 via SSLG4 OPSC-106: yield 9700/MeV, $\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns, ABSLENGTH=120 cm (verified effective MPT inputs).
- ▶ Broadcom AFBR-S4N66P024M: PDE 63% at 420 nm.

Scope

Top 70 SiPM is a simulation coverage study. Real hardware: 32 Top SiPMs. All reported timing is intrinsic, without SPTR or electronics jitter.

EndTop geometry and channel map

- ▶ EJ-230 bar: $1400 \times 60 \times 10 \text{ mm}^3$.
- ▶ End L IDs 0–7; End R IDs 8–15.
- ▶ Top L IDs 16–50: $-692, -672, \dots, -12 \text{ mm}$.
- ▶ Top R IDs 51–85: $+12, +32, \dots, +692 \text{ mm}$.
- ▶ Each Top element and window: $6 \times 6 \text{ mm}^2$.

Exact constraints

First centers are 5 mm from each bar end. Pitch is 20 mm except the deliberate central $-12/+12 \text{ mm}$ pair, whose pitch is 24 mm.

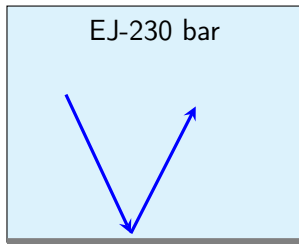
Instrumented and wrapped faces

Face	Optical state	Readout
$-X, +X$	open	8 End SiPMs each
$+Y$	Mylar with 70 exact windows	70 Top SiPMs
$-Y, -Z, +Z$	complete Mylar	none

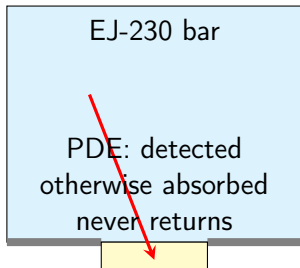
External layer

Black Tedlar is deliberately not modeled; it only blocks ambient light. The non-reflected Mylar fraction is already absorbed in the optical model.

Window optics: why timing and collection change



Mylar, fixed $R = 0.98$: reflected photons return



index-matched window

Consequence

Every window is an optical drain. Long, multiply reflected paths have more chances to meet one, so late End photons are preferentially removed.

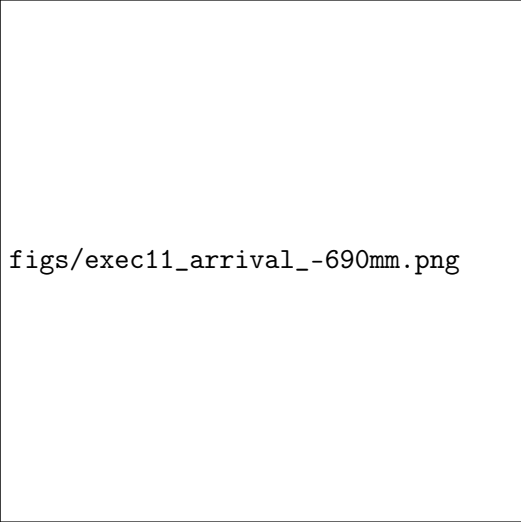
Position $x=-690$ mm

figs/muon_-690mm_geometry.png

figs/muon_-690mm_top_profile.png

Quantity		Value
End L N_{pe}		3363.1 ± 18.1
End R N_{pe}		41.1 ± 0.3
Nearest Top ID	16 (window-track exception)	
Nearest Top N_{pe}		350.0 ± 2.2

Run 1 | $x = -690$ mm | Photon arrival $\langle N(t) \rangle$

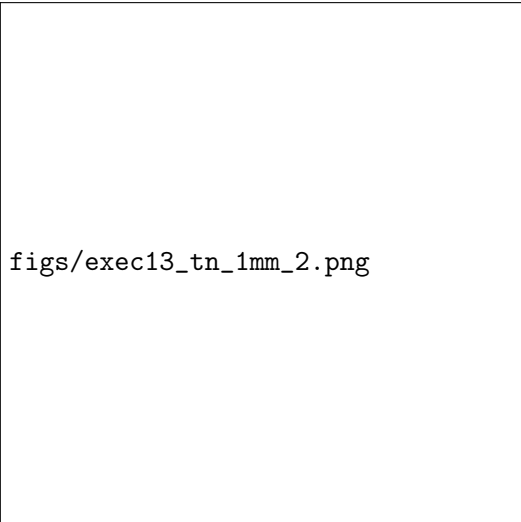


`figs/exec11_arrival_-690mm.png`

Run 1 | $x = -690$ mm | Time-to-threshold t_N

`figs/exec13_tN_-690mm.png`

Run 1 | $x = -690$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

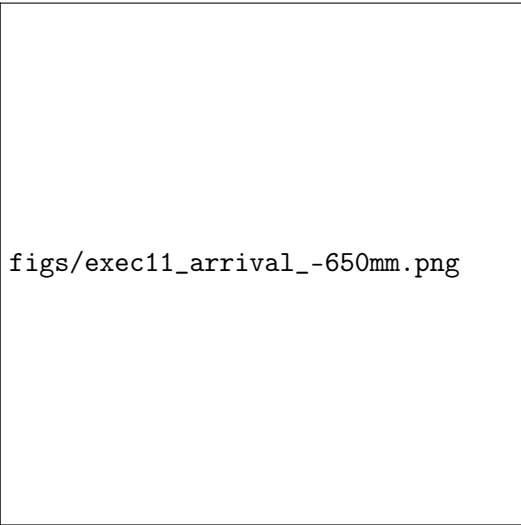
Position $x=-650$ mm

figs/muon_-650mm_geometry.png

figs/muon_-650mm_top_profile.png

Quantity	Value
End L N_{pe}	2366.2 ± 13.1
End R N_{pe}	44.3 ± 0.3
Nearest Top ID	18 (window-track exception)
Nearest Top N	251.0 ± 2.2

Run 2 | $x = -650$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-650mm.png`

Run 2 | $x = -650$ mm | Time-to-threshold t_N



`figs/exec13_tN_-650mm.png`

Run 2 | $x = -650$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

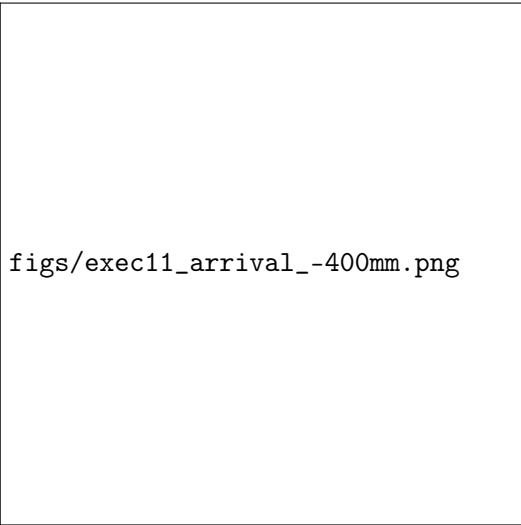
Position $x=-400$ mm

figs/muon_-400mm_geometry.png

figs/muon_-400mm_top_profile.png

Quantity	Value
End L N_{pe}	691.3 ± 3.9
End R N_{pe}	74.5 ± 0.5
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	411.3 ± 2.2

Run 3 | $x = -400$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-400mm.png`

Run 3 | $x = -400$ mm | Time-to-threshold t_N

`figs/exec13_tN_-400mm.png`

Run 3 | $x = -400$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

Position $x=0$ mm

figs/muon_0mm_geometry.png

figs/muon_0mm_top_profile.png

Quantity	Value
End L N_{pe}	194.4 ± 1.1
End R N_{pe}	195.2 ± 1.1
Nearest Top ID	51 (exact geometry)
Nearest Top N_{pe}	197.3 ± 2.2

Run 4 | $x = +0$ mm | Photon arrival $\langle N(t) \rangle$

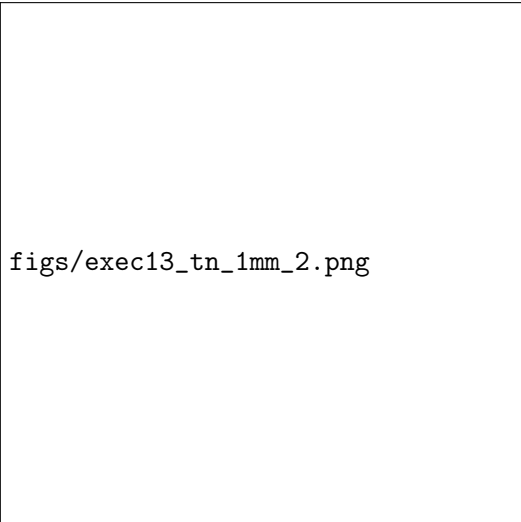


`figs/exec11_arrival_0mm.png`

Run 4 | $x = +0$ mm | Time-to-threshold t_N

`figs/exec13_tN_0mm.png`

Run 4 | $x = +0$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

Position $x=400$ mm

figs/muon_400mm_geometry.png

figs/muon_400mm_top_profile.png

Quantity	Value
End L N_{pe}	75.0 ± 0.5
End R N_{pe}	692.5 ± 4.0
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	411.4 ± 2.1

Run 5 | $x = +400$ mm | Photon arrival $\langle N(t) \rangle$

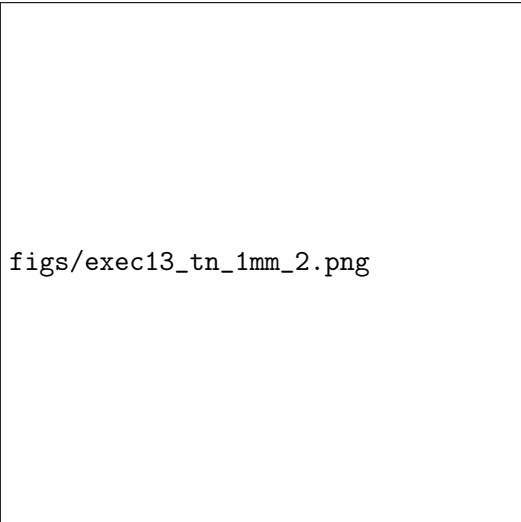


`figs/exec11_arrival_400mm.png`

Run 5 | $x = +400$ mm | Time-to-threshold t_N

`figs/exec13_tN_400mm.png`

Run 5 | $x = +400$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

Position $x=650$ mm

figs/muon_650mm_geometry.png

figs/muon_650mm_top_profile.png

Quantity	Value
End L N_{pe}	44.4 ± 0.3
End R N_{pe}	2357.1 ± 13.0
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	250.0 ± 2.1

Run 6 | $x = +650$ mm | Photon arrival $\langle N(t) \rangle$



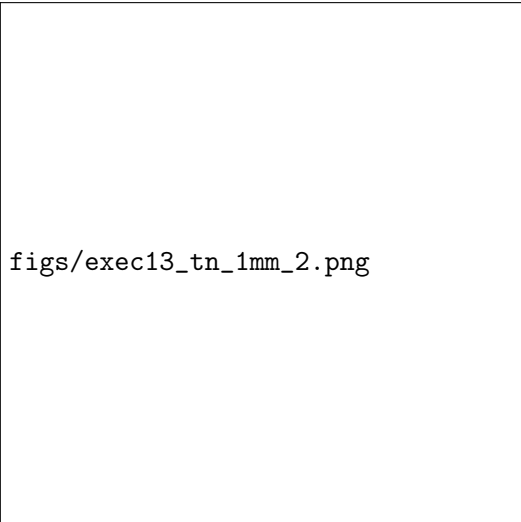
`figs/exec11_arrival_650mm.png`

Run 6 | $x = +650$ mm | Time-to-threshold t_N



`figs/exec13_tN_650mm.png`

Run 6 | $x = +650$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

Position $x=690$ mm

figs/muon_690mm_geometry.png

figs/muon_690mm_top_profile.png

Quantity	Value
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End L N_{pe}	41.4 ± 0.3
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End R N_{pe}	3383.2 ± 18.3
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Nearest Top ID	85 (window-track exception)
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Nearest Top N_{pe}	252.0 ± 2.2
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Run 7 | $x = +690$ mm | Photon arrival $\langle N(t) \rangle$



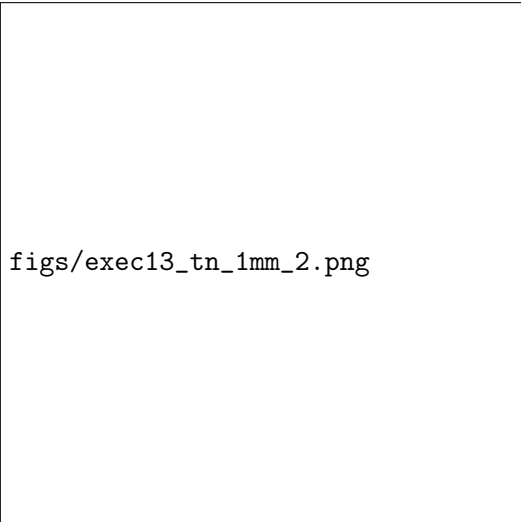
`figs/exec11_arrival_690mm.png`

Run 7 | $x = +690$ mm | Time-to-threshold t_N

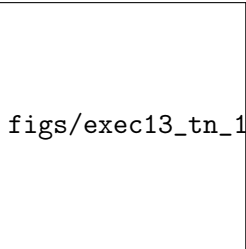


`figs/exec13_tN_690mm.png`

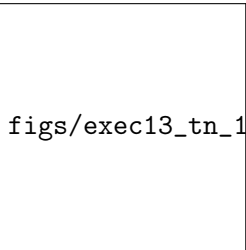
Run 7 | $x = +690$ mm | $\langle t_n \rangle$ per channel (dispersion)



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png



figs/exec13_tn_1mm_2.png

Error bars = RMS (event-to-event). Dashed band: statistical floor $\sigma_{stat}(t_n) \approx \sqrt{n} \tau_d / \langle N_{pe} \rangle$ ($\tau_d = 1.5$ ns). See dispersion-decomposition frame for physical interpretation.

$\sigma(t_N)$ vs beam position: synthesis



figs/exec13_tN_summary.png

Interpretation and connection to σ_{group}

$\sigma(t_N)$ is the single-side timing resolution at an ideal digital threshold of N detected PE; it connects the cumulative $\langle N(t) \rangle$ curves to the published σ_{group} values. At high N_{pe} ($x=\pm 690, \pm 650, \pm 400$ near their respective End), $\sigma(t_4) \sim 15$ ps is the pure photon-counting limit; $\sigma_{\text{group}} \sim 50$ ps is larger because the analog SUM4 discriminator adds jitter from the SPE pulse convolution ($\tau_r = 0.5$ ns). At $x = 0$ (moderate N_{pe} , both Ends equidistant), photon-counting noise dominates and both converge to ~ 120 ps.

Why does t_n disperse? Decomposition into four mechanisms

1. **Order statistics on the emission tail (irreducible):** t_n are order statistics of N_{pe} draws from the emission pdf ($\tau_r = 0.5$ ns, $\tau_d = 1.5$ ns); successive gaps $\sim \text{Exp}(\tau_d / (N_{pe} - n + 1))$, $\sigma_{\text{stat}}(t_n) \approx \sqrt{n} \tau_d / N_{pe}$ for $n \ll N_{pe}$ — grows monotonically with n even with perfect optics.
2. **Landau N_{pe} fluctuations:**
 $\text{Var}_L(t_n) = (\partial \langle t_n \rangle / \partial N_{pe})^2 \text{Var}(N_{pe})$ with $\text{Var}(N_{pe})$ from the documented $F = 1 + c \langle N_{pe} \rangle$. Statistical but non-Poissonian.
3. **Optical path-length dispersion:** TIR photons at $n = 1.58$ ($\theta_c \approx 39$) span axial speeds $c / (n \cos \theta) \rightarrow 1.4\text{--}1.5$ ns/m per metre plus Mylar bounce tail ($R = 0.98$). Low- n photons: quasi-direct; high- n : reflection-dominated $\Rightarrow \text{RMS} \propto n$.
4. **Attenuation truncation:** distant channels ($\lambda_{\text{eff}} \approx 33$ cm) populate only $n = 1\text{--}3$ with large purely statistical bars.

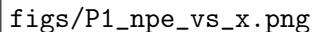
Takeaway

$\text{RMS}(t_n) - \sqrt{n} \tau_d / N_{pe}$ isolates the optical-propagation contribution — the per-channel diagnostic in the $\langle t_n \rangle$ dispersion frames and the same dispersion that limits σ_{group} .

Simulation only / intrinsic

Top: simulation prediction, no test-beam counterpart. End: test-beam comparable, intrinsic precision (no SPTR / no jitter).

Photon budget versus beam position



figs/P1_npe_vs_x.png

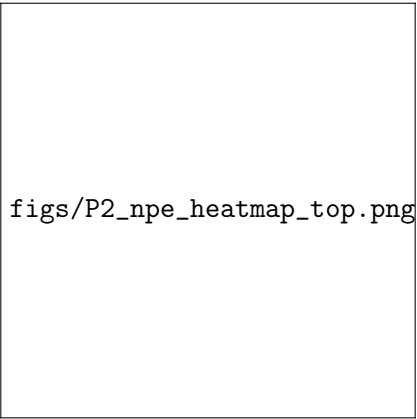
How to read this plot

Why: Quantify N_{pe} collected at each End and nearest-Top channel as the muon track scans the bar.

Axes: x: beam position [mm]; y: mean N_{pe} per event (End sums left/right and nearest Top).

Takeaway: N_{pe} decreases exponentially with track-to-End distance ($\lambda_{eff} \approx 33 \text{ cm} \ll \text{ABSLENGTH} = 120 \text{ cm}$, system effective length); nearest-Top N_{pe} peaks directly under the SiPM window.

Top localization: full coverage diagonal



figs/P2_npe_heatmap_top.png

How to read this plot

Why: Verify that the highest- N_{pe} Top channel is always the geometrically nearest one across all 31 positions.

Axes: x: beam position [mm]; y: Top channel ID; colour: mean N_{pe} .

Takeaway: Strict diagonal coverage; window-track exception positions ($x \equiv 10 \pmod{20}$ mm) are labelled and pass the $< 15\%$ deficit criterion.

Attenuation and the historical mean-time slope

figs/fits_attenuation_velocity.png

$\lambda_{eff,L} = 30.70 \pm 0.02$ cm, $\lambda_{eff,R} = 30.82 \pm 0.02$ cm; the mean-time slope is tail-biased and is not a propagation velocity.

How to read this plot

Why: Measure the effective attenuation length λ_{eff} and test whether the mean-time slope encodes a light-propagation velocity.

Axes: Left: mean N_{pe} vs track-to-End distance [cm]; exponential fit gives λ_{eff} . Right: mean arrival time [ns] vs same distance.

Takeaway: $\lambda_{eff} \approx 31$ cm $\ll$ ABSLENGTH = 120 cm: λ_{eff} is a system effective length combining bulk absorption, Mylar reflection losses, and Top window drainage—not the material property. The mean-time slope is tail-biased; no propagation velocity is reliably extracted.

Why the Fano factor grows with $\langle N_{pe} \rangle$

1. **Pure Poisson (fixed ΔE):** $\text{Var}(N_{pe}) = \langle N_{pe} \rangle$, hence $F \equiv 1$.
2. **Landau fluctuations in ΔE** (law of total variance):

$$\text{Var}(N_{pe}) = \langle N_{pe} \rangle + \left(\frac{\langle N_{pe} \rangle}{\langle \Delta E \rangle} \right)^2 \text{Var}(\Delta E),$$

$\Rightarrow F = 1 + c \langle N_{pe} \rangle$, $c = \text{Var}(\Delta E) / \langle \Delta E \rangle^2$
(collection-efficiency terms absorbed in c).

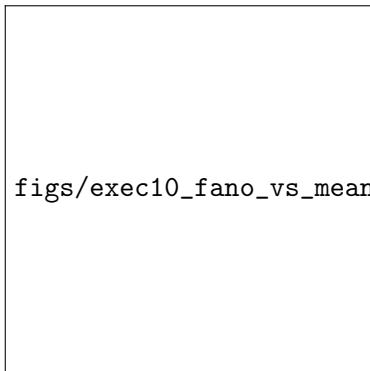
3. **Poisson limit $F \rightarrow 1$:** only at $\langle N_{pe} \rangle \rightarrow 0$
(counting noise dominates over energy-loss fluctuations).
4. **Effective relative fluctuation:** $\sqrt{c} \approx 0.249$ is the combined energy-loss + collection jitter, consistent with a thin (10 mm) plastic absorber.

Fit result (see next slide)

$$c = 0.062547 \pm 0.000117$$

The *same* Landau ΔE fluctuation that shapes the asymmetric N_{pe} spectrum (next section) also generates the super-Poissonian Fano factor: **one physical mechanism, two observables.**

Landau dominance: Fano factor versus mean N_{pe}



$F = 1 + c\langle N_{pe} \rangle$, $c = 0.062547 \pm 0.000117$, $\chi^2/ndf = 1.41$. Poisson $F = 1$ is approached only at low light.

How to read this plot

Why: Measure $F = \text{Var}(N_{pe})/\langle N_{pe} \rangle$ to distinguish Poisson ($F = 1$) from Landau-dominated fluctuations.

Axes: x: $\langle N_{pe} \rangle$ per event; y: $\text{Var}/\langle N_{pe} \rangle$. Dashed: Poisson $F = 1$. Line: linear fit $F = 1 + c\langle N_{pe} \rangle$.

Takeaway: F grows linearly: Landau ΔE fluctuations dominate counting noise at all light levels. $\sqrt{c}$ is the effective relative energy-loss fluctuation of this 10 mm absorber.

Why Landau (and why Moyal as analytic proxy)

Why not Gaussian:

- ▶ In 10 mm of plastic the muon undergoes many collisions, but the energy-transfer cross-section ($d\sigma/d\varepsilon \propto 1/\varepsilon^2$, Rutherford/Møller) has a hard tail.
- ▶ Rare large transfers (δ -rays) dominate the variance; the variance per collision diverges and the Central Limit Theorem does not apply.
- ▶ Result: asymmetric distribution with a hard right tail — the Landau distribution.
- ▶ Gaussian (Vavilov→Gauss) limit only applies to *thick* absorbers.

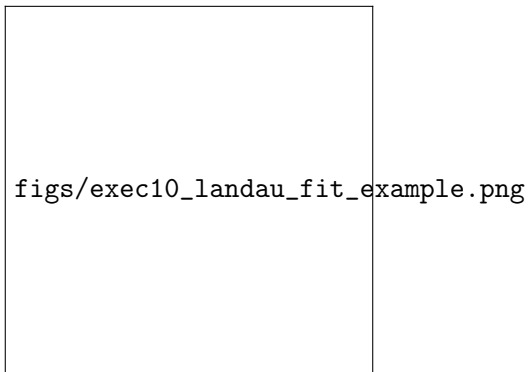
Why Moyal:

- ▶ Landau density has no closed form (contour integral).
- ▶ The Moyal function is a closed-form analytic approximation to the Landau core and early tail, enabling stable MINUIT fits.
- ▶ *Known limitation*: extreme Moyal tail undershoots true Landau tail; fit quality degrades at high N_{pe} .

Moyal $\otimes$ Gauss:

- ▶ Photo-electron counting (Poisson) and collection fluctuations broaden the peak symmetrically.
- ▶ Full model: Moyal $\otimes$ Gaussian.
- ▶ Connects to Fano slide: same Landau ΔE fluctuation, two observables (spectrum shape + super-Poissonian F).

High-light example: Moyal–Gaussian Landau proxy



Moyal approximates Landau. Some channels have poor χ^2 or an unidentifiable Gaussian width; all recorded in `exec10_landau_mpv.csv`.

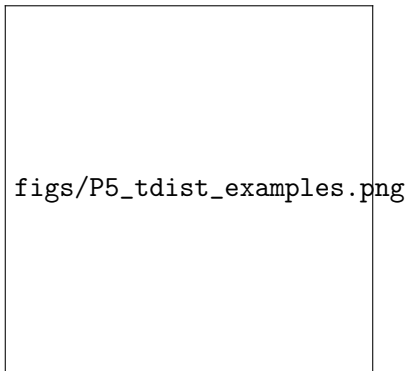
How to read this plot

Why: Fit the N_{pe} spectrum at a high-yield position to quantify the Landau MPV and width.

Axes: x: N_{pe} per event; y: counts. Data (step), Moyal⊗Gauss fit (curve).

Takeaway: Core described by Moyal proxy; hard right tail from δ -rays. MPVs are future inputs for test-beam ToT calibration.

Arrival-time examples: absolute rate and normalized shape



Left: entries per event per ns. Right: area-normalized distributions. FPT = first detected photon time per event.

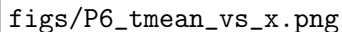
How to read this plot

Why: Illustrate the arrival-time structure of individual Top channels and contrast absolute rate vs shape.

Axes: x: arrival time [ns]; y: left is entries/event/ns, right is area-normalized density [1/ns].

Takeaway: FPT distribution captures the prompt scintillation leading edge; the broad tail reflects multiple-reflection photons.

Mean arrival time and FPT versus x



figs/P6_tmean_vs_x.png

How to read this plot

Why: Track how the mean photon arrival time and FPT scale with beam position.

Axes: x: beam position [mm]; y: mean arrival time [ns] and mean FPT [ns].

Takeaway: Both estimators increase with track-to-End distance; the mean is more tail-biased than FPT and yields a larger apparent slope.

Nearest-Top Landau/Moyal MPVs at key positions

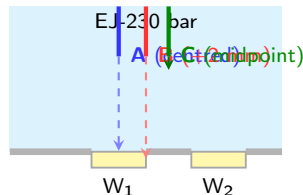
x [mm]	channel	MPV [Npe]	Landau width [Npe]	χ^2/ndf
-690	16	275.0 ± 0.9	23.3 ± 0.8	1.61
-650	18	275.6 ± 0.9	23.7 ± 0.8	1.58
-400	31	317.5 ± 1.0	26.8 ± 0.9	1.56
+0	50	318.5 ± 1.0	26.0 ± 0.9	1.17
+400	70	320.1 ± 1.0	25.3 ± 0.8	1.23
+650	83	275.0 ± 0.9	24.7 ± 0.8	1.87
+690	85	278.1 ± 0.9	24.6 ± 0.8	2.15

Interpretation

The right tail originates primarily from Landau-like energy-loss fluctuations; Poisson counting dominates only in the few-PE limit. MPVs are future inputs for the pending test-beam ToT(NPE) calibration comparison.

Window-track collection effect: mechanism

- ▶ Top SiPMs couple through index-matched windows ($6 \times 6 \text{ mm}^2$) in the +Y Mylar face.
- ▶ First-pass photon collection depends on the solid angle subtended by the window from the muon track (geometric working hypothesis, consistent with five-run pattern).
- ▶ Empirically: a track at $\pm 2 \text{ mm}$ from the window centre collects $\approx 9.8\%$ more N_{pe} ($\approx 12\sigma$); null at centre and at the midpoint; symmetric in mirror (C2).
- ▶ Scan positions with $x \equiv 10 \pmod{20} \text{ mm}$ fall at $\pm 2 \text{ mm}$ from a window centre — hence the “(window-track exception)” label in the tables.
- ▶ Effect is local, symmetric, characterised and bounded: localization diagonal and channel map remain valid.



Consequence for the scan

Track B subtends a larger first-pass solid angle to W_1 ; fewer multiple reflections $\Rightarrow$ higher N_{pe} at the nearest channel. At midpoint (C) both windows contribute equally and the effect is null.

Window-track collection effect: five-run test

Run	contrast [Npe]	significance
Existing -650	34.60 ± 2.94	11.78
A -652 center	2.86 ± 2.65	1.08
B -642 midpoint	-0.33 ± 3.35	-0.10
C1 -648 (+4 mm)	21.08 ± 3.20	6.59
C2 -654 mirror	34.38 ± 2.95	11.64

`figs/exec08b_window_dip_profiles.png`

Intrinsic End SUM4 timing



figs/P7_deltaT_end.png

$\sigma_{group} = \sigma(\Delta T_{AB})/\sqrt{2}$; no SPTR/jitter. The 88 ps hardware context is not directly comparable.

How to read this plot

Why: Measure the intrinsic Same-End SUM4 coincidence timing resolution without SPTR or electronics jitter.

Axes: x: beam position [mm]; y: σ_{group} [ps]. Lines for left and right End groups.

Takeaway: σ_{group} decreases toward the nearer End (higher N_{pe} , earlier FPT); consistent with photon-counting scaling $\propto 1/\sqrt{N_{pe}}$.

End arrival-time tails without any time acceptance window: narrower EndTop is physical

x	side	EndTop [ps]	End-only [ps]	ratio
-400	L	50.0 ± 0.8	66.1 ± 0.5	0.756 ± 0.013
-400	R	219.9 ± 3.5	248.4 ± 1.8	0.885 ± 0.015
+0	L	107.5 ± 1.7	131.4 ± 0.9	0.818 ± 0.014
+0	R	107.2 ± 1.7	129.5 ± 0.9	0.828 ± 0.014
+400	L	223.8 ± 3.5	247.5 ± 1.8	0.904 ± 0.016
+400	R	51.5 ± 0.8	65.9 ± 0.5	0.781 ± 0.014

Confirmed mechanism

Across $x = 0, \pm 400$ mm, EndTop lowers t_{99} by 1.14 ns and suppresses late fractions at more than 10.3σ .

figs/exec09_tail_comparison.png

No time acceptance window

All End-photon arrivals are used without a $t < t_{max}$ cut (exec08b_timing_gate.py, exec09_timing_mechanism.py). EndTop tail narrowing persists over the full range: physical, not an artefact.

Apparent timing slopes: no propagation estimator passes

`figs/exec10_velocity_estimators.png`

Estimator	slope-derived [cm/ns]	χ^2/ndf
all-photon mean	25.616 ± 0.005	3189.9
all-photon t_{50}	25.625 ± 0.005	3615.1
mean FPT	26.758 ± 0.003	1002.6

Refuted predictions

Mean FPT is not close to $c/n \simeq 19$ cm/ns. The fixed-threshold $f(t > 10 \text{ ns})$ increases with distance. Every global line fit is rejected.

How to read this plot

Why: Test whether any timing estimator reveals a light-propagation velocity.

Axes: x: track-to-End distance [cm]; y: timing estimator [ns]. Fits shown.

Takeaway: No estimator yields a statistically acceptable propagation-velocity fit; the slope is dominated by the long late-tail bias.

Top readout: analytic timing estimate only

x [mm]	N_{pe}	$\sigma_{t,est}$ [ps]
-690	350.9 ± 2.0	59.92 ± 0.17
-650	351.0 ± 2.0	59.91 ± 0.17
-400	411.3 ± 2.3	55.35 ± 0.16
+0	407.3 ± 2.2	55.62 ± 0.15
+400	411.4 ± 2.4	55.34 ± 0.16
+650	350.9 ± 2.1	59.93 ± 0.18
+690	353.0 ± 2.0	59.75 ± 0.17

Definition, derivation, and scope

$$\sigma_{t,est} = \sqrt{\tau_r \tau_d} / \sqrt{N_{pe}} = 1.122 \text{ ns} / \sqrt{N_{pe}}$$

Threshold at low N_{pe} , double-exponential emission (Knoll, ch. 8; $\tau_r = 0.5 \text{ ns}$, $\tau_d = 1.5 \text{ ns}$). **Analytic orientation only**; Top has no test-beam counterpart.

$\sigma_{t,est}$ vs. simulated $\sigma(t_4)$

x	$\sigma_{t,est}$ [ps]	$\sigma(t_4)$ [ps]
-690	60 ± 0	9
-650	60 ± 0	8
-400	55 ± 0	8
+0	56 ± 0	8
+400	55 ± 0	8
+650	60 ± 0	9
+690	60 ± 0	8

$\sigma(t_4)$: photon-counting lower bound.

Numerical conclusions

- ▶ At center: End sum 302.1 ± 1.2 , nearest Top 366.0 ± 2.0 ; at $x = -690$ mm: End_L 3073.3 ± 16.8 , nearest Top 318.7 ± 1.8 .
- ▶ λ_{eff} : L 30.70 ± 0.02 cm, R 30.82 ± 0.02 cm ($\ll$ ABSLENGTH = 120 cm; system effective length, not material property).
- ▶ Nearest-Top $F = 1 + (0.062547 \pm 0.000117)\langle N_{pe} \rangle$ ($\chi^2/ndf = 1.41$): Landau energy-loss fluctuations dominate. Same mechanism: asymmetric N_{pe} spectrum (Moyal $\otimes$ Gauss) and $F > 1$.
- ▶ Intrinsic End σ_{group} : mean 226 ps, range 12–595 ps.
- ▶ EndTop/End-only timing ratio: N/A for EJ-230 (End-only runs not available; see H00K_EXEC09_EJ230); no time-acceptance window in any script.
- ▶ $\sigma(t_4)$ (near-End SUM4): 15–123 ps (high- N_{pe} to low); pure photon-counting lower bound, factor ~ 3 below σ_{group} at high N_{pe} due to SPE pulse-convolution jitter in the analog discriminator. Nearest-Top $\sigma(t_4)$: 8–9 ps.
- ▶ Four findings: window-track mechanism defined and characterised; late-tail drain confirmed; complete 86-channel coverage and localisation validated; t_N photon-counting timing quantified.

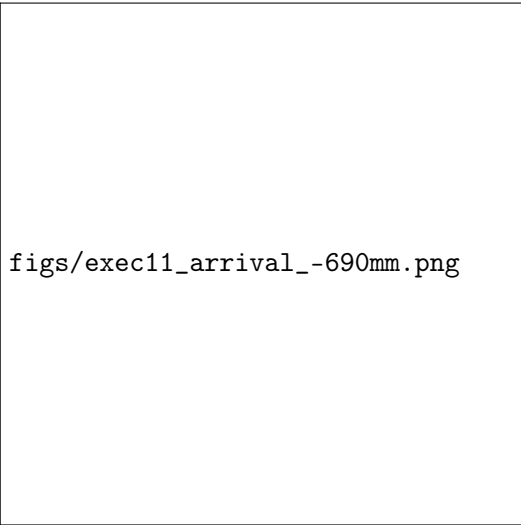
Position $x=-690$ mm

figs/muon_-690mm_geometry.png

figs/muon_-690mm_top_profile.png

Quantity	Value
End L N_{pe}	3363.1 ± 18.1
End R N_{pe}	41.1 ± 0.3
Nearest Top ID	16 (window-track exception)
Nearest Top N	250.0 ± 2.2

Run 1 | $x = -690$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-690mm.png`

Position $x=-670$ mm

figs/muon_-670mm_geometry.png

figs/muon_-670mm_top_profile.png

Quantity	Value
End L N_{pe}	2764.3 ± 15.0
End R N_{pe}	42.6 ± 0.3
Nearest Top ID	17 (window-track exception)
Nearest Top N_{pe}	250.4 ± 2.2

Run 2 | $x = -670$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-670mm.png`

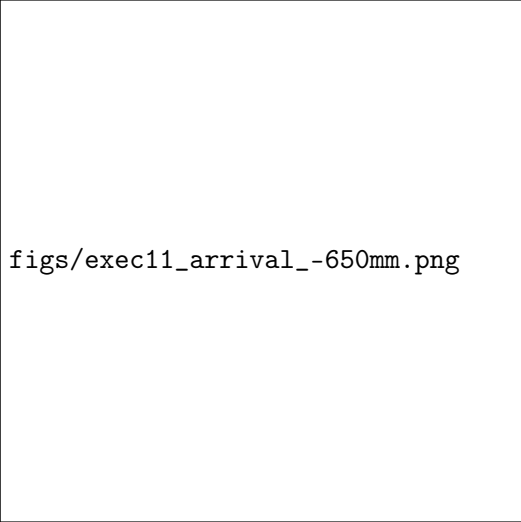
Position $x=-650$ mm

figs/muon_-650mm_geometry.png

figs/muon_-650mm_top_profile.png

Quantity	Value
End L N_{pe}	2366.2 ± 13.1
End R N_{pe}	44.3 ± 0.3
Nearest Top ID	18 (window-track exception)
Nearest Top N_{pe}	251.0 ± 2.2

Run 3 | $x = -650$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-650mm.png`

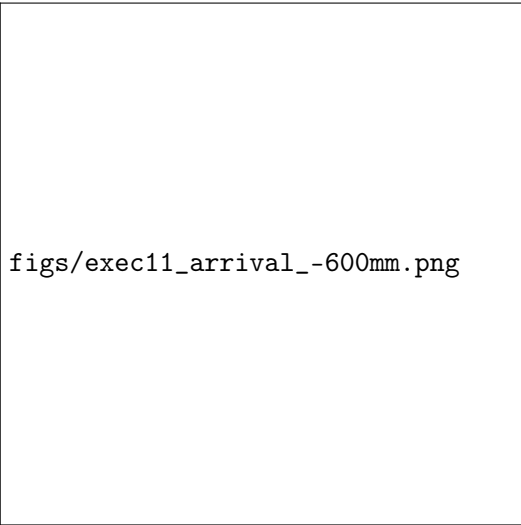
Position $x=-600$ mm

figs/muon_-600mm_geometry.png

figs/muon_-600mm_top_profile.png

Quantity	Value
End L N_{pe}	1673.6 ± 9.2
End R N_{pe}	48.0 ± 0.3
Nearest Top ID	21 (exact geometry)
Nearest Top N_{pe}	422.0 ± 2.2

Run 4 | $x = -600$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-600mm.png`

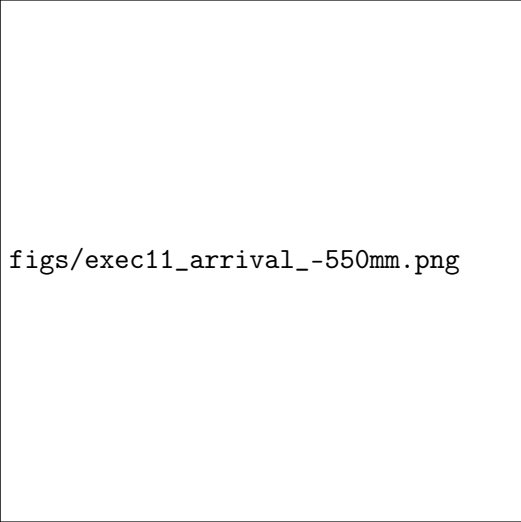
Position $x=-550$ mm

figs/muon_-550mm_geometry.png

figs/muon_-550mm_top_profile.png

Quantity	Value
End L N_{pe}	1323.9 ± 7.6
End R N_{pe}	55.1 ± 0.4
Nearest Top ID	23 (window-track exception)
Nearest Top N_{pe}	252.0 ± 2.1

Run 5 | $x = -550$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-550mm.png`

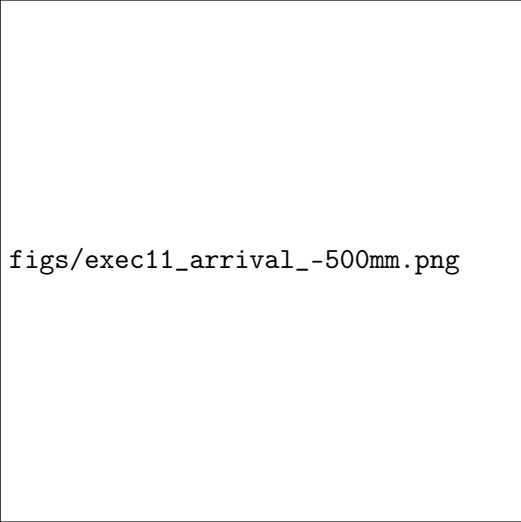
Position $x=-500$ mm

figs/muon_-500mm_geometry.png

figs/muon_-500mm_top_profile.png

Quantity	Value
End L N_{pe}	1042.2 ± 5.9
End R N_{pe}	59.8 ± 0.4
Nearest Top ID	26 (exact geometry)
Nearest Top N	419.9 ± 2.2

Run 6 | $x = -500$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-500mm.png`

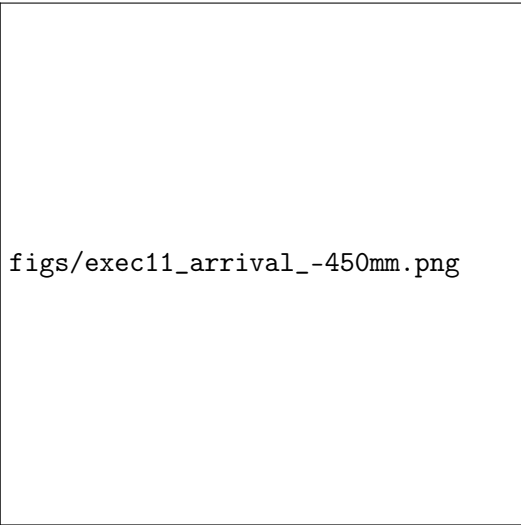
Position $x=-450$ mm

figs/muon_-450mm_geometry.png

figs/muon_-450mm_top_profile.png

Quantity	Value
End L N_{pe}	832.9 ± 4.2
End R N_{pe}	65.8 ± 0.4
Nearest Top ID	28 (window-track exception)
Nearest Top N_{pe}	245.9 ± 1.2

Run 7 | $x = -450$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-450mm.png`

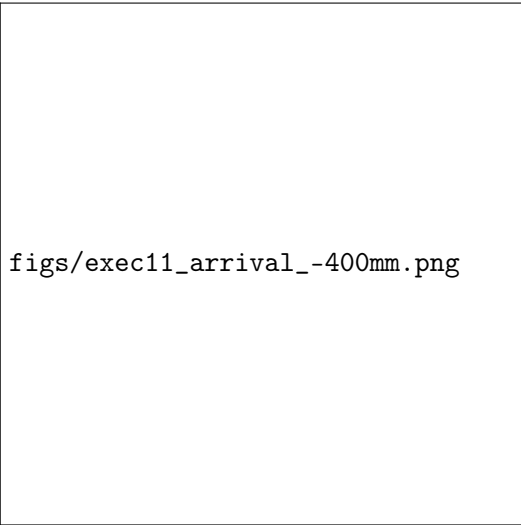
Position $x=-400$ mm

figs/muon_-400mm_geometry.png

figs/muon_-400mm_top_profile.png

Quantity	Value
End L N_{pe}	691.3 ± 3.9
End R N_{pe}	74.5 ± 0.5
Nearest Top ID	31 (exact geometry)
Nearest Top N_{pe}	411.3 ± 2.2

Run 8 | $x = -400$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-400mm.png`

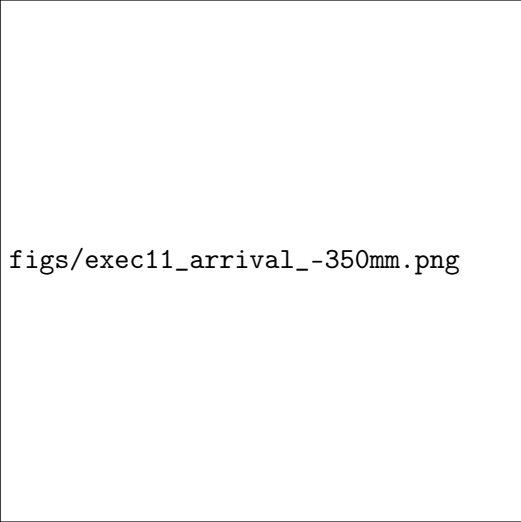
Position $x=-350$ mm

figs/muon_-350mm_geometry.png

figs/muon_-350mm_top_profile.png

Quantity		Value
End L N_{pe}		575.0 ± 3.1
End R N_{pe}		84.1 ± 0.5
Nearest Top ID	33 (window-track exception)	
Nearest Top N_{pe}		249.2 ± 1.2

Run 9 | $x = -350$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-350mm.png`

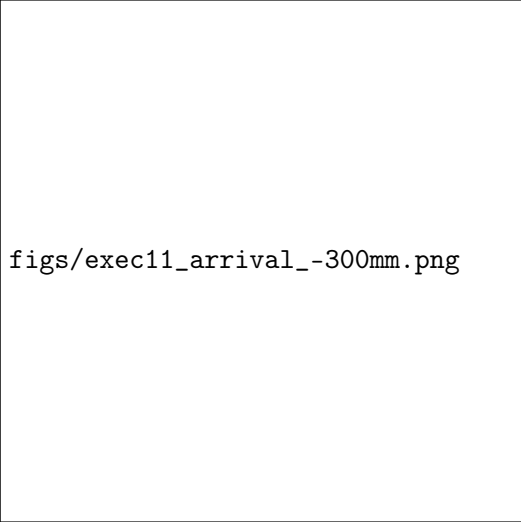
Position $x=-300$ mm

figs/muon_-300mm_geometry.png

figs/muon_-300mm_top_profile.png

Quantity	Value
End L N_{pe}	482.3 ± 2.8
End R N_{pe}	94.2 ± 0.6
Nearest Top ID	36 (exact geometry)
Nearest Top N	412.7 ± 2.5

Run 10 | $x = -300$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-300mm.png`

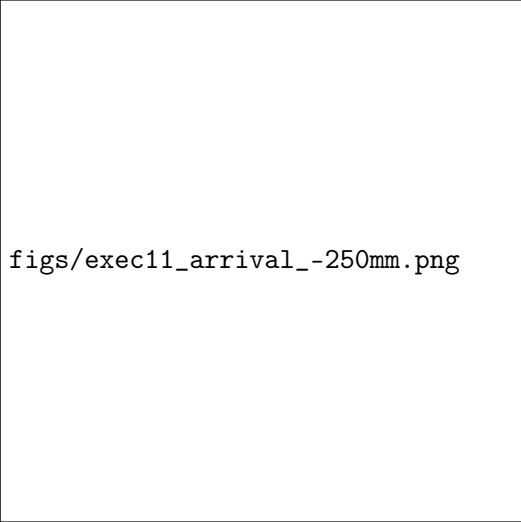
Position $x=-250$ mm

figs/muon_-250mm_geometry.png

figs/muon_-250mm_top_profile.png

Quantity		Value
End L N_{pe}		415.8 ± 2.4
End R N_{pe}		105.5 ± 0.7
Nearest Top ID	38 (window-track exception)	
Nearest Top N_{pe}		252.2 ± 2.1

Run 11 | $x = -250$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-250mm.png`

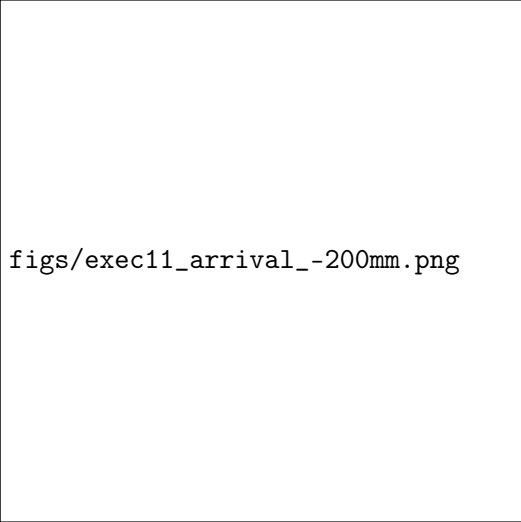
Position $x=-200$ mm

figs/muon_-200mm_geometry.png

figs/muon_-200mm_top_profile.png

Quantity	Value
End L N_{pe}	353.5 ± 2.0
End R N_{pe}	118.4 ± 0.7
Nearest Top ID	41 (exact geometry)
Nearest Top N_{pe}	412.5 ± 2.2

Run 12 | $x = -200$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-200mm.png`

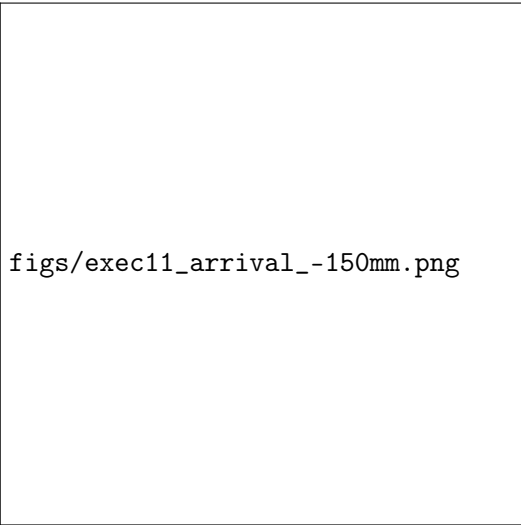
Position $x=-150$ mm

figs/muon_-150mm_geometry.png

figs/muon_-150mm_top_profile.png

Quantity		Value
End L N_{pe}		302.1 ± 1.8
End R N_{pe}		134.7 ± 0.8
Nearest Top ID	43 (window-track exception)	
Nearest Top N_{pe}		250.0 ± 2.1

Run 13 | $x = -150$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-150mm.png`

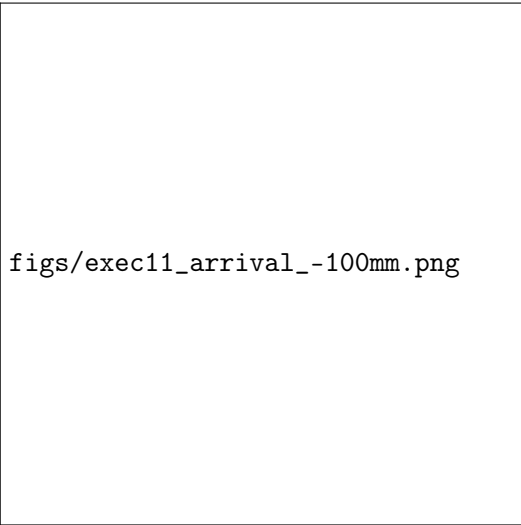
Position $x=-100$ mm

figs/muon_-100mm_geometry.png

figs/muon_-100mm_top_profile.png

Quantity	Value
End L N_{pe}	256.3 ± 1.5
End R N_{pe}	151.4 ± 0.9
Nearest Top ID	46 (exact geometry)
Nearest Top N_{pe}	411.0 ± 2.2

Run 14 | $x = -100$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-100mm.png`

Position $x=-50$ mm

figs/muon_-50mm_geometry.png

figs/muon_-50mm_top_profile.png

Quantity		Value
End L N_{pe}		226.7 ± 1.4
End R N_{pe}		172.7 ± 1.1
Nearest Top ID	48 (window-track exception)	
Nearest Top N_{pe}		252.0 ± 2.1

Run 15 | $x = -50$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_-50mm.png`

Position $x=0$ mm

figs/muon_0mm_geometry.png

figs/muon_0mm_top_profile.png

Quantity	Value
End L N_{pe}	194.4 ± 1.1
End R N_{pe}	195.2 ± 1.1
Nearest Top ID	51 (exact geometry)
Nearest Top N_{pe}	197.3 ± 2.2

Run 16 | $x = +0$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_0mm.png`

Position $x=50$ mm

figs/muon_50mm_geometry.png

figs/muon_50mm_top_profile.png

Quantity	Value
End L N_{pe}	171.6 ± 1.1
End R N_{pe}	226.4 ± 1.3
Nearest Top ID	53 (window-track exception)
Nearest Top N_{pe}	242.2 ± 2.1

Run 17 | $x = +50$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_50mm.png`

Position $x=100$ mm

figs/muon_100mm_geometry.png

figs/muon_100mm_top_profile.png

Quantity	Value
End L N_{pe}	152.2 ± 0.8
End R N_{pe}	258.3 ± 1.4
Nearest Top ID	55 (exact geometry)
Nearest Top N_{pe}	412.8 ± 2.2

Run 18 | $x = +100$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_100mm.png`

Position $x=150$ mm

figs/muon_150mm_geometry.png

figs/muon_150mm_top_profile.png

Quantity		Value
End L N_{pe}		135.5 ± 0.8
End R N_{pe}		303.6 ± 1.7
Nearest Top ID	58 (window-track exception)	
Nearest Top N_{pe}		252.1 ± 2.2

Run 19 | $x = +150$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_150mm.png`

Position $x=200$ mm

figs/muon_200mm_geometry.png

figs/muon_200mm_top_profile.png

Quantity	Value
End L N_{pe}	118.5 ± 0.7
End R N_{pe}	354.1 ± 2.1
Nearest Top ID	60 (exact geometry)
Nearest Top N_{pe}	412.3 ± 2.1

Run 20 | $x = +200$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_200mm.png`

Position $x=250$ mm

figs/muon_250mm_geometry.png

figs/muon_250mm_top_profile.png

Quantity	Value
End L N_{pe}	104.8 ± 0.7
End R N_{pe}	417.0 ± 2.4
Nearest Top ID	63 (window-track exception)
Nearest Top N_{pe}	252.0 ± 2.1

Run 21 | $x = +250$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_250mm.png`

Position $x=300$ mm

figs/muon_300mm_geometry.png

figs/muon_300mm_top_profile.png

Quantity		Value
End L N_{pe}		92.5 ± 0.6
End R N_{pe}		477.5 ± 2.8
Nearest Top ID	65 (statistical tie: two nearest)	
Nearest Top N_{pe}		499.7 ± 2.2

Run 22 | $x = +300$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_300mm.png`

Position $x=350$ mm

figs/muon_350mm_geometry.png

figs/muon_350mm_top_profile.png

Quantity	Value
End L N_{pe}	84.0 ± 0.5
End R N_{pe}	575.5 ± 3.3
Nearest Top ID	68 (window-track exception)
Nearest Top N_{pe}	240.0 ± 2.2

Run 23 | $x = +350$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_350mm.png`

Position $x=400$ mm

figs/muon_400mm_geometry.png

figs/muon_400mm_top_profile.png

Quantity	Value
End L N_{pe}	75.0 ± 0.5
End R N_{pe}	692.5 ± 4.0
Nearest Top ID	70 (exact geometry)
Nearest Top N_{pe}	411.4 ± 2.1

Run 24 | $x = +400$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_400mm.png`

Position $x=450$ mm

figs/muon_450mm_geometry.png

figs/muon_450mm_top_profile.png

Quantity	Value
End L N_{pe}	67.1 ± 0.4
End R N_{pe}	845.1 ± 4.8
Nearest Top ID	73 (window-track exception)
Nearest Top N_{pe}	252.2 ± 2.1

Run 25 | $x = +450$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_450mm.png`

Position $x=500$ mm

figs/muon_500mm_geometry.png

figs/muon_500mm_top_profile.png

Quantity	Value
End L N_{pe}	59.9 ± 0.4
End R N_{pe}	1039.9 ± 5.9
Nearest Top ID	75 (exact geometry)
Nearest Top N_{pe}	419.5 ± 2.2

Run 26 | $x = +500$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_500mm.png`

Position $x=550$ mm

figs/muon_550mm_geometry.png

figs/muon_550mm_top_profile.png

Quantity	Value
End L N_{pe}	55.1 ± 0.4
End R N_{pe}	1325.9 ± 7.7
Nearest Top ID	78 (window-track exception)
Nearest Top N_{pe}	252.3 ± 2.1

Run 27 | $x = +550$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_550mm.png`

Position $x=600$ mm

figs/muon_600mm_geometry.png

figs/muon_600mm_top_profile.png

Quantity	Value
End L N_{pe}	48.6 ± 0.3
End R N_{pe}	1687.6 ± 9.3
Nearest Top ID	80 (exact geometry)
Nearest Top N_{pe}	411.6 ± 2.2

Run 28 | $x = +600$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_600mm.png`

Position $x=650$ mm

figs/muon_650mm_geometry.png

figs/muon_650mm_top_profile.png

Quantity	Value
End L N_{pe}	44.4 ± 0.3
End R N_{pe}	2357.1 ± 13.0
Nearest Top ID	83 (window-track exception)
Nearest Top N_{pe}	250.0 ± 2.1

Run 29 | $x = +650$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_650mm.png`

Position $x=670$ mm

figs/muon_670mm_geometry.png

figs/muon_670mm_top_profile.png

Quantity	Value
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End L N_{pe}	42.2 ± 0.3
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End R N_{pe}	2763.5 ± 15.3
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Nearest Top ID	84 (window-track exception)
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Nearest Top N_{pe}	250.2 ± 2.2
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Run 30 | $x = +670$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_670mm.png`

Position $x=690$ mm

figs/muon_690mm_geometry.png

figs/muon_690mm_top_profile.png

Quantity	Value
End L N_{pe}	41.4 ± 0.3
End R N_{pe}	3383.2 ± 18.3
Nearest Top ID	85 (window-track exception)
Nearest Top N_{pe}	252.0 ± 2.2

Run 31 | $x = +690$ mm | Photon arrival $\langle N(t) \rangle$



`figs/exec11_arrival_690mm.png`

Backup: validation and gates

- ▶ 31/31 ROOT files pass: event IDs 0–1999, matching beam x , aggregate IDs 0–85.
- ▶ Top localization passes all positions: strict nearest maximum except the documented $x \equiv 10 \pmod{20}$ window-track effect (maximum among nearest two, deficit below 15%).
- ▶ At $x=300$ mm the two candidates differ by only 0.08 sigma and pass as a statistical tie.
- ▶ MPT checks pass: RINDEX 1.58, yield 9700/MeV, decay 1.5 ns, rise 0.5 ns, ABSLENGTH 120 cm, emission peak ≈ 391 nm.

Backup: ID18 impact maps

`figs/exec08b_id18_impact_maps.png`

Backup: why a 4-PE leading-edge threshold

1. **Noise rejection at hardware parity:** 1 PE is dark-count territory; 2–3 PE is reachable by correlated crosstalk in the real SiPM. 4 PE is the lowest fixed threshold outside the correlated-noise spectrum, matching the operating point of a hardware leading-edge discriminator.
2. **Slope optimum:** $\sigma_t \approx \sigma_V / (dV/dt)$; with the 0.5/5 ns SPE the SUM4 pulse has maximal rising-edge slope in the first few PE. Lower thresholds expose the high relative variance of first-photon order statistics; higher thresholds reduce slope and increase walk.
3. **Far-end efficiency:** minimum signal in the scan is End R $\langle N_{pe} \rangle = 41.1 \pm 0.3$ at $x = -690$ mm; 4 PE ($\approx 10\%$ of the minimum pulse) keeps trigger efficiency $\approx 100\%$ at all 31 positions, avoiding position-dependent selection bias.
4. **Bounded, correctable walk:** fixed-threshold time-walk is small compared to σ_{group} at 4 PE and is parametrized for future correction via HOOK_WALK (no correction applied in any campaign).

Backup: timing methodology

- ▶ SUM4 groups: $\{0..3\}$, $\{4..7\}$, $\{8..11\}$, $\{12..15\}$.
- ▶ Single-PE response: normalized 0.5/5 ns double exponential.
- ▶ Absolute leading-edge threshold: 4 PE equivalent (noise rejection, slope optimum, $\approx 100\%$ efficiency at all 31 positions; see dedicated backup slide for full rationale).
- ▶ Estimator: $\sigma(\Delta T_{AB})/\sqrt{2}$.
- ▶ HOOK_WALK: pending; no time-walk correction exists or was invented.
- ▶ EXEC_09 anti-artifact audit verifies common threshold, pulse, $t=0$ and estimator.

Backup: glossary

FPT	First photon time: first detected photon per event.	SPTR	Single Photon Time Resolution: SiPM intrinsic single-PE jitter. Excluded from all EXEC_12 results (jitter= 0); re-introduced via EXEC_02b hooks, summed in quadrature with electronics jitter.
Fano factor	$F \equiv \text{Var}(N_{pe}) / \langle N_{pe} \rangle = 1 + c \langle N_{pe} \rangle$ when Landau ΔE fluctuations are present.	SUM4	Analog sum of four SiPMs: {0–3}, {4–7}, {8–11}, {12–15} (congruent_sum4_timing.C: 218–221).
Group A / Group B	Same-End SUM4 groups: A={0–3}, B={4–7} on End L; mirrors {8–11}, {12–15} on End R. $\sigma_{\text{group}} = \sigma(\Delta T_{AB}) / \sqrt{2}$ (sum4_timing.C:218–221).	Timing gate	exec08b_timing_gate.py tests EndTop vs. End-only σ_{group} at $x = 0, \pm 400$ mm; no time acceptance cut on photons.
HOOK_WALK	Reserved time-walk correction point; no correction in any campaign (sum4_timing.C:305–315).	Window-track effect	± 2 mm track offset from window centre increases first-pass solid angle; empirically $\approx 9.8\%$ at $\approx 12\sigma$; null at centre and midpoint. Geometric working hypothesis.
Intrinsic sigma	Excludes SPTR and electronics jitter; 88 ps hardware value not directly comparable.	t_N / time-to-threshold	N -th detected photoelectron arrival time per event. $\sigma(t_N)$: photon-counting timing resolution at ideal N -PE digital threshold.
Moyal	Closed-form Landau approximation for stable MINUIT fits. Undershoots extreme tail.	(exact geometry)	Nearest SiPM = maximum, or window-track exception; $x = 300$ mm is a statistical tie.

Backup: EndTop versus End-only tail audit



`figs/exec09_tail_comparison.png`